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Fundamentals of Statistical Learning

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Contents

1	Crash Course	1
1.1	Why Probability?	1
1.2	Experiments, Outcomes, and Events	1
1.3	Operations on Events	2
1.4	Assigning Probabilities	3

1

Crash Course

1.1 Why Probability?

Observed data are the results of processes whose behavior we may not fully know. To reason about them, we describe the process with a probabilistic model. Probability lets us work forward from a model to possible outcomes; statistical inference uses observed outcomes to learn about the model.

Example 1: A coin toss

Let $\theta = P(\text{heads})$ be the probability that a coin lands heads. If θ is known, we can ask how likely it is to observe three heads in five tosses. If instead we observe three heads in five tosses while θ is unknown, we can use those data to learn about θ .

1.2 Experiments, Outcomes, and Events

Before assigning probabilities, we must specify what the experiment can produce. We start with a finite set of possible results.

Definition 1: Outcomes and events

A random experiment is a process whose result is not known in advance. Each possible result is an *outcome* ω . The set of all possible outcomes is the *sample space* Ω . An *event* is a subset $A \subseteq \Omega$; it occurs when the observed outcome belongs to A .

Example 2: Rolling a die

For one roll of a six-sided die, the sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event “the result is even” is $A = \{2, 4, 6\}$. The whole sample space Ω is the certain event, while $\emptyset$ is the impossible event.

1.3 Operations on Events

Events are sets, so set operations let us describe more complex events. For $A, B \subseteq \Omega$, the union $A \cup B$ occurs when at least one of A and B occurs. The intersection $A \cap B$ occurs when both occur. The complement $A^c = \Omega \setminus A$ occurs when A does not. Two events are *disjoint* if $A \cap B = \emptyset$.

Example 3: Combining die events

Keep $A = \{2, 4, 6\}$ from Example 2, and let $B = \{4, 5, 6\}$ be the event “the result is at least four”. Then

$$A \cup B = \{2, 4, 5, 6\}, \quad A \cap B = \{4, 6\}, \quad A^c = \{1, 3, 5\}.$$

Proposition 1: De Morgan’s laws

For any events A and B ,

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

Proof. Fix an outcome $\omega \in \Omega$. For the first identity, check when ω belongs to the left-hand side:

$$\begin{aligned} \omega \in (A \cup B)^c &\iff \omega \notin A \cup B \\ &\iff \omega \notin A \text{ and } \omega \notin B \\ &\iff \omega \in A^c \cap B^c. \end{aligned}$$

The same reasoning for the second identity gives

$$\begin{aligned} \omega \in (A \cap B)^c &\iff \omega \notin A \cap B \\ &\iff \omega \notin A \text{ or } \omega \notin B \\ &\iff \omega \in A^c \cup B^c. \end{aligned}$$

Each pair of sets has the same members, so the identities follow. □

1.4 Assigning Probabilities

Definition 2: Probability on a finite sample space

A probability measure P assigns a number $P(A)$ to every event $A \subseteq \Omega$ such that:

1. $0 \leq P(A) \leq 1$ for every event A ;
2. $P(\Omega) = 1$;
3. if $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

The last property is called *finite additivity*.

Because A and A^c are disjoint and their union is Ω , these axioms give $P(A^c) = 1 - P(A)$. In particular, $P(\emptyset) = 0$.

Example 4: A fair die

If the six outcomes of a die roll are equally likely, each has probability $1/6$. The event $A = \{2, 4, 6\}$ therefore has probability $P(A) = 3/6 = 1/2$.